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MILANO 1863

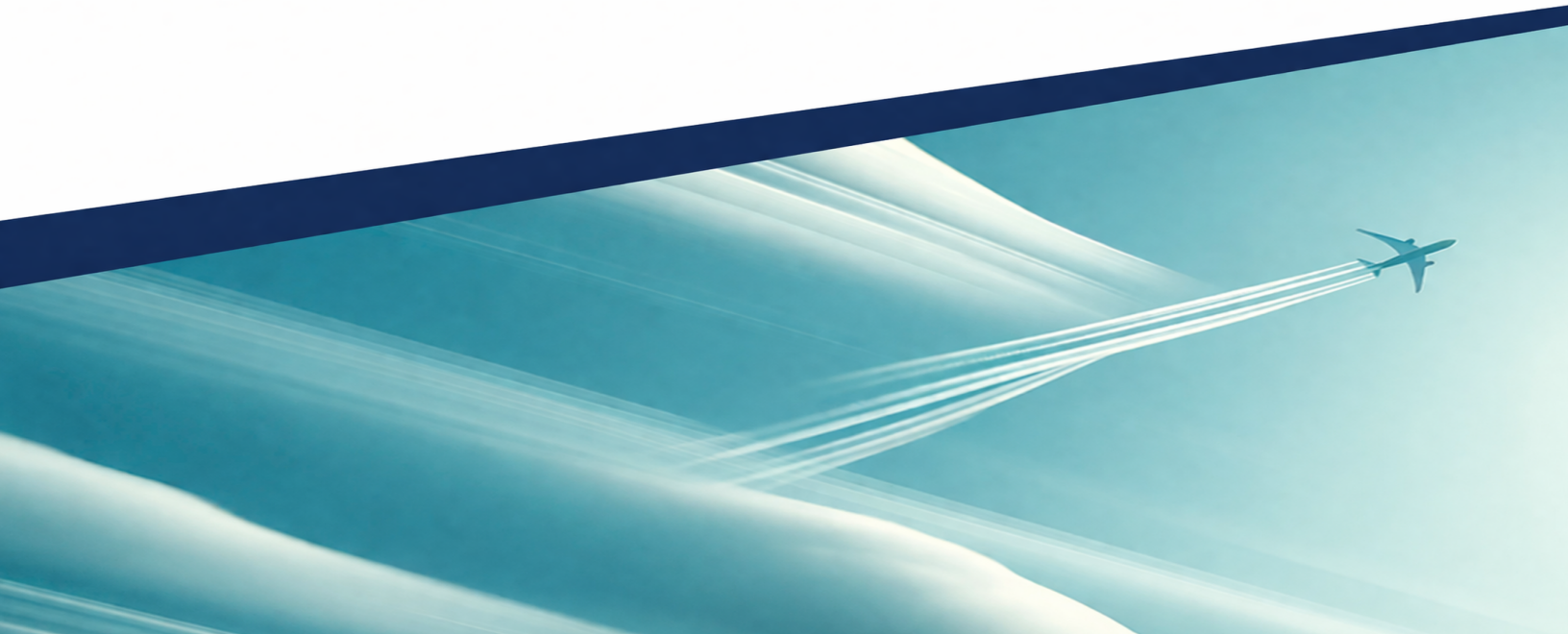
ASSIGNMENT 1

STRUCTURAL DYNAMICS AND AEROELASTICITY

Analytical Report

POLITECNICO DI MILANO
Laurea Magistrale – Aeronautical Engineering

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Problem Setup

The wing has two DOFs — bending $w(y)$ and torsion $\theta(y)$ — with a strut at $y = y_B$, aerodynamic center offset e from the elastic axis, and a tip aileron of span b_a and deflection β . The Ritz–Galerkin method is applied to perform a static aeroelastic analysis.

Divergence Dynamic Pressure q_D - Derivation

Shape Functions

Root clamp imposes $w(0) = w'(0) = \theta(0) = 0$. The Ritz ansatz $w = N_w(y) \phi_w$, $\theta = N_\theta(y) \phi_\theta$ requires N_w quadratic and N_θ linear. Applying BCs and normalizing to unity at the tip :

$$\boxed{N_w(y) = \left(\frac{y}{b}\right)^2, \quad N_\theta(y) = \frac{y}{b}} \quad (1)$$

Principle of Virtual Work

In order to compute q_D , we need to find the equations of motion. To do so, we use the PVW : We need to compute the internal and external virtual work.

Internal Virtual Work

Substituting $N_w'' = 2/b^2$ and $N_\theta' = 1/b$:

$$\delta W_i = \int_0^b \delta w_{,yy} EJ w_{,yy} dy + \int_0^b \delta \theta_{,y} GJ \theta_{,y} dy = \delta \phi_w \frac{4EJ}{b^3} \phi_w + \delta \phi_\theta \frac{GJ}{b} \phi_\theta \quad (2)$$

$$\boxed{\delta W_i = \delta \phi^T \begin{bmatrix} K_w & 0 \\ 0 & K_\theta \end{bmatrix} \phi, \quad K_w = \frac{4EJ}{b^3}, \quad K_\theta = \frac{GJ}{b}} \quad (3)$$

External Virtual Work

$$\delta W_e = \delta W_{\text{AERO}}^{\text{wing}} + \delta W_{\text{AERO}}^{\text{aileron}} + \delta W_{\text{STRUT}} + \delta W_{\text{INERTIA}} + \delta W_{\text{GRAVITY}}$$

Since we are interested in the divergence dynamic pressure q_D , we can neglect inertia and gravity (static case, no external forces). Thus :

$$\boxed{\delta W_e = \delta W_A^w + \delta W_A^a + \delta W_{\text{STR}}}$$

(1) Wing aerodynamic virtual work δW_A^w The kinematically admissible AC displacement is $h_{AC} = w + e\theta$, and strip lift reads $L'_w = q c C_{L/\alpha} \theta$ (perturbation, $\alpha_{\text{eff}} = \theta$, $C_{L_0} = 0$). Thus :

$$\delta W_A^w = \int_0^b \delta h_{AC} L'_w dy = \int_0^b \delta \phi_w \left(\frac{y}{b}\right)^2 q c C_{L/\alpha} \frac{y}{b} \phi_\theta dy + \int_0^b \delta \phi_\theta \frac{y}{b} e q c C_{L/\alpha} \frac{y}{b} \phi_\theta dy \quad (4)$$

$$\boxed{\delta W_A^w = \delta \phi^T q \mathbf{K}_A \phi}, \quad \mathbf{K}_A = \begin{bmatrix} 0 & \frac{cb}{4} C_{L/\alpha} \\ 0 & \frac{ecb}{3} C_{L/\alpha} \end{bmatrix} \quad (5)$$

$\mathbf{K}_A$ is the aerodynamic stiffness matrix.

(2) Aileron aerodynamic virtual work δW_A^a Not needed for q_D (homogeneous problem) but derived for later use. The aileron spans $[b - b_a, b]$ with deflection β ; perturbation gives $C_{m_{AC}} = C_{m/\beta} \beta$. The virtual work splits into a lift contribution and a pitching moment contribution :

$$\delta W_A^a = \int_{b-b_a}^b \delta(w + e\theta) q c C_{L/\beta} \beta dy + \int_{b-b_a}^b \delta\theta q c^2 C_{m/\beta} \beta dy \quad (6)$$

$$\boxed{\delta W_A^a = \delta \phi^T q \beta \mathbf{F}_a}, \quad \mathbf{F}_a = \frac{c}{b} \left\{ \begin{array}{c} C_{L/\beta} \cdot \frac{b^3 - (b - b_a)^3}{3b} \\ (e C_{L/\beta} + c C_{m/\beta}) \frac{b^2 - (b - b_a)^2}{2} \end{array} \right\} = \begin{Bmatrix} F_{a1} \\ F_{a2} \end{Bmatrix} \quad (7)$$

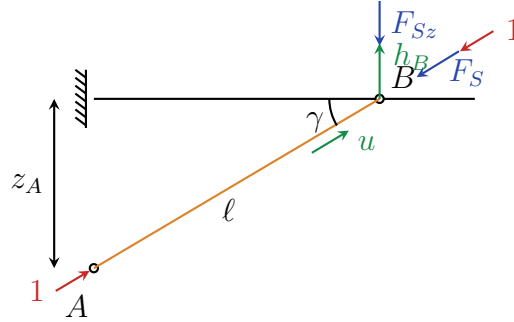


FIGURE 1 – Equivalent model of the strut, replaced by an equivalent axial spring of stiffness K_{STR} .

(3) Strut virtual work δW_{STR} The strut axial force and its z -component follow from geometry ($\ell = z_A / \sin \gamma$, $u = h_B \sin \gamma$) :

$$\boxed{F_{Sz} = - \underbrace{\frac{EA}{z_A}}_{K_{STR}} \sin^3 \gamma h_B} \quad (8)$$

The displacement at attachment point B, using $\eta_B = z_A / (b \tan \gamma)$:

$$h_B = w(y_B) - x_B \theta(y_B) = \eta_B [\eta_B \phi_w - x_B \phi_\theta]$$

Substituting into $\delta W_{STR} = \delta h_B (-K_{STR}) h_B$:

$$\boxed{\delta W_{STR} = -\delta \phi^T K_{STR} \eta_B^2 \begin{bmatrix} \eta_B^2 & -\eta_B x_B \\ -\eta_B x_B & x_B^2 \end{bmatrix} \phi} \quad (9)$$

Application of PVW : $\delta W_i = \delta W_e$

$$\delta \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix}^T \begin{bmatrix} K_w & 0 \\ 0 & K_\theta \end{bmatrix} \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} = \delta \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix}^T \left(q \mathbf{K}_A \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} + q \beta \mathbf{F}_a - K_{\text{STR}} \eta_B^2 \begin{bmatrix} \eta_B^2 & -\eta_B x_B \\ -\eta_B x_B & x_B^2 \end{bmatrix} \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} \right) \quad (10)$$

Assembling all contributions and invoking the arbitrariness of $\delta \phi$ (unforced case $\beta = 0$ for q_D computation) :

$$\left(\begin{bmatrix} K_w + K_{\text{STR}} \eta_B^4 & -K_{\text{STR}} \eta_B^3 x_B \\ -K_{\text{STR}} \eta_B^3 x_B & K_\theta + K_{\text{STR}} \eta_B^2 x_B^2 \end{bmatrix} - q \mathbf{K}_A \right) \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad (11)$$

This is $(\mathbf{K}_S - q_D \mathbf{K}_A) \phi = \mathbf{0}$, with :

$$\mathbf{K}_S = \begin{bmatrix} K_{S_1} & K_{S_{12}} \\ K_{S_{12}} & K_{S_2} \end{bmatrix}, \quad \mathbf{K}_A = \begin{bmatrix} 0 & \mathbf{K}_{A_1} = \frac{cb}{4} C_{L/\alpha} \\ 0 & \mathbf{K}_{A_2} = \frac{ecb}{3} C_{L/\alpha} \end{bmatrix} \quad (12)$$

$$K_{S_1} = \frac{4EJ}{b^3} + \frac{EA z_A^3}{b^4} \frac{\cos^4 \gamma}{\sin \gamma}, \quad K_{S_{12}} = -\frac{EA z_A^2}{b^3} \cos^3 \gamma x_B, \quad K_{S_2} = \frac{GJ}{b} + \frac{EA \sin \gamma \cos^2 \gamma z_A x_B^2}{b^2} \quad (13)$$

For divergence, the condition is : $\det(\mathbf{K}_S - q_D \mathbf{K}_A) = 0$. Expanding the determinant :

$$\det(\mathbf{K}_S - q_D \mathbf{K}_A) = K_{S_1} K_{S_2} - q_D K_{S_1} K_{A_2} - K_{S_{12}}^2 + q_D K_{S_{12}} K_{A_1} = 0 \quad \Rightarrow \quad q_D = \frac{K_{S_1} K_{S_2} - (K_{S_{12}})^2}{K_{S_1} K_{A_2} - K_{S_{12}} K_{A_1}} \quad (14)$$

Control Effectiveness Computation

The control effectiveness is defined as $E_C = \frac{L}{L_R}$. The total lift is obtained by spanwise integration :

$$L = \int_0^b L'(y) dy = qc \left(\int_0^b C_{L/\alpha} \underbrace{\theta(y)}_{\frac{y}{b} \phi_\theta} dy + \int_{b-b_a}^b C_{L/\beta} \beta dy \right)$$

$$\boxed{L = \underbrace{qcb_a C_{L/\beta} \beta}_{L_R \text{ (Rigid wing)}} + \underbrace{qc \frac{b C_{L/\alpha}}{2} \phi_\theta}_{L_E \text{ (Elastic torsion)}}} \quad (15)$$

We need to determine ϕ_θ . From the first question, we have :

$$(\mathbf{K}_S - q \mathbf{K}_A) \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} = q \beta \mathbf{F}_a \quad \text{where we define } \mathbf{K}_{\text{TOT}} = \mathbf{K}_S - q \mathbf{K}_A \quad (16)$$

For $q < q_D$, $\mathbf{K}_{\text{TOT}}$ is invertible :

$$\mathbf{K}_{\text{TOT}}^{-1} = \frac{1}{\det(\mathbf{K}_{\text{TOT}})} \begin{bmatrix} K_{\text{TOT},22} & -K_{\text{TOT},12} \\ -K_{\text{TOT},21} & K_{\text{TOT},11} \end{bmatrix} \quad (17)$$

Where $\det(\mathbf{K}_{\text{TOT}})$ is defined at equation (14). Thus :

$$\begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} = q\beta \mathbf{K}_{\text{TOT}}^{-1} \mathbf{F}_a \quad (18)$$

The aileron influence load vector $\mathbf{F}_a$ is defined at (7). From matrix multiplication :

$$\phi_\theta = \frac{q\beta}{\det(\mathbf{K}_{\text{TOT}})} (K_{\text{TOT},11} F_{a2} - K_{\text{TOT},21} F_{a1}) \quad (19)$$

Substituting into the lift expression :

$$L = qc\beta \left(b_a C_{L/\beta} + \frac{b C_{L/\alpha}}{2} \cdot \frac{q}{\det(\mathbf{K}_{\text{TOT}})} (K_{\text{TOT},11} F_{a2} - K_{\text{TOT},21} F_{a1}) \right) \quad (20)$$

Dividing by L_R gives the control effectiveness E_C :

$$E_C = 1 + q \cdot \frac{b}{b_a} \cdot \frac{C_{L/\alpha}}{C_{L/\beta}} \cdot \frac{1}{2 \det(\mathbf{K}_{\text{TOT}})} \cdot (K_{\text{TOT},11} F_{a2} - K_{\text{TOT},21} F_{a1}) \quad (21)$$

The expression of E_C separates the rigid contribution from the aeroelastic correction. At $q = 0$, one recovers $E_C = 1$ (rigid limit); as q increases, aileron-induced torsion modifies total lift. The correction is amplified as $\det(\mathbf{K}_{\text{TOT}}) \rightarrow 0$ (divergence), so E_C may vanish before divergence – defining the control reversal pressure.

Wing deformed shape – Strut effect

Compare wing deformation for $\beta_0 = 10^\circ$ at $q = 0.4 q_D^{(0)}$, where $q_D^{(0)}$ is the divergence dynamic pressure without strut.

Displacement field.

$$z(x, y) = w(y) - x \theta(y) = \left(\frac{y}{b} \right)^2 \phi_w - x \frac{y}{b} \phi_\theta \quad (22)$$

Aeroelastic equilibrium. Inverting $\mathbf{K}_{\text{TOT}} \{\phi_w, \phi_\theta\}^\top = q\beta \mathbf{F}_a$ gives

$$\phi_w = \frac{q\beta}{\det \mathbf{K}_{\text{TOT}}} (K_{\text{TOT},22} F_{a1} - K_{\text{TOT},12} F_{a2}), \quad (23)$$

$$\phi_\theta = \frac{q\beta}{\det \mathbf{K}_{\text{TOT}}} (K_{\text{TOT},11} F_{a2} - K_{\text{TOT},21} F_{a1}), \quad (24)$$

hence

$$z(x, y) = \frac{q\beta}{\det \mathbf{K}_{\text{TOT}}} \frac{y}{b} \left[\frac{y}{b} (K_{\text{TOT},22} F_{a1} - K_{\text{TOT},12} F_{a2}) - x (K_{\text{TOT},11} F_{a2} - K_{\text{TOT},21} F_{a1}) \right]. \quad (25)$$

Configurations compared.

- **With strut** : full structural model ;
- **Without strut** : strut axial stiffness set to $EA = 0$.

To achieve the comparison we define :

$$\delta_z = 100 \frac{|z_{\text{strut}} - z_{\text{ns}}|}{|z_{\text{ns}}|}. \quad (26)$$

Which is the relative deviation of the strut configuration with respect to the no-strut reference.

Convergence of q_D w.r.t. Ritz shape functions number

To study convergence of the Ritz approximation, bending and torsional displacements are expanded on a hierarchical polynomial basis that satisfies the clamped-root essentials conditions $w(0) = w'(0) = \theta(0) = 0$:

$$w(y) = \sum_{i=1}^N \left(\frac{y}{b}\right)^{i+1} \phi_{w_i} = \underline{\mathbf{N}}_w^T \underline{\phi}_w, \quad \underline{\mathbf{N}}_w = \{\eta^{i+1}\}_N, \quad \underline{\phi}_w = \{\phi_{w_i}\}_N, \quad \eta = \frac{y}{b} \quad (27)$$

$$\theta(y) = \sum_{i=1}^N \left(\frac{y}{b}\right)^i \phi_{\theta_i} = \underline{\mathbf{N}}_\theta^T \underline{\phi}_\theta, \quad \underline{\mathbf{N}}_\theta = \{\eta^i\}_N, \quad \underline{\phi}_\theta = \{\phi_{\theta_i}\}_N. \quad (28)$$

Internal virtual work

Substituting the Ritz ansatz into $\delta W_i = \int_0^b \delta w'' EJ w'' dy + \int_0^b \delta \theta' GJ \theta' dy$ and factoring out the generalized DOFs :

$$\delta W_i = \delta \underline{\Phi}^T \underline{\mathbf{K}}_{\text{WSTIF}}^{(N)} \underline{\Phi}, \quad \underline{\mathbf{K}}_{\text{WSTIF}}^{(N)} = \begin{bmatrix} \underline{\mathbf{K}}_w^{(N)} & \underline{\mathbf{0}} \\ \underline{\mathbf{0}} & \underline{\mathbf{K}}_\theta^{(N)} \end{bmatrix}, \quad \underline{\Phi} = \left\{ \begin{matrix} \underline{\phi}_w \\ \underline{\phi}_\theta \end{matrix} \right\}, \quad (29)$$

where the stiffness sub-matrices are

$$\underline{\mathbf{K}}_w^{(N)} = \int_0^b \underline{\mathbf{N}}_w'' EJ \underline{\mathbf{N}}_w''^T dy, \quad \underline{\mathbf{K}}_\theta^{(N)} = \int_0^b \underline{\mathbf{N}}_\theta' GJ \underline{\mathbf{N}}_\theta'^T dy, \quad (30)$$

with

$$\underline{\mathbf{N}}_w'' = \frac{1}{b^2} \{(i+1)i\eta^{i-1}\}_N, \quad \underline{\mathbf{N}}_\theta' = \frac{1}{b} \{i\eta^{i-1}\}_N. \quad (31)$$

Aerodynamic virtual work

The AC displacement is $h_{AC} = \underline{\mathbf{N}}_w^T \underline{\phi}_w + e \underline{\mathbf{N}}_\theta^T \underline{\phi}_\theta$ and strip lift reads $L' = qcC_{L/\alpha} \underline{\mathbf{N}}_\theta^T \underline{\phi}_\theta$. Expanding $\delta W_{e,A}^w = \int_0^b \delta h_{AC} L' dy$ gives :

$$\delta W_A^w = \delta \underline{\Phi}^T q \underline{\mathbf{K}}_A^{(N)} \underline{\Phi}, \quad \underline{\mathbf{K}}_A^{(N)} = \begin{bmatrix} \underline{\mathbf{0}} & \underline{\mathbf{K}}_{A1}^{(N)} \\ \underline{\mathbf{0}} & \underline{\mathbf{K}}_{A2}^{(N)} \end{bmatrix}, \quad (32)$$

$$\underline{\mathbf{K}}_{A1}^{(N)} = \int_0^b \underline{\mathbf{N}}_w cC_{L/\alpha} \underline{\mathbf{N}}_\theta^T dy, \quad \underline{\mathbf{K}}_{A2}^{(N)} = \int_0^b \underline{\mathbf{N}}_\theta e cC_{L/\alpha} \underline{\mathbf{N}}_\theta^T dy. \quad (33)$$

Strut contribution

The strut stiffness $K_{\text{STR}} = (EA/z_A) \sin^3 \gamma$ is unchanged. The attachment-point displacement

$$h_B = \underline{\mathbf{N}}_w^T(\eta_B) \underline{\phi}_w - x_B \underline{\mathbf{N}}_\theta^T(\eta_B) \underline{\phi}_\theta, \quad \eta_B = \frac{z_A}{b \tan \gamma}, \quad (34)$$

enters $\delta W_{\text{STR}} = -\delta h_B K_{\text{STR}} h_B$. Substituting the Ritz approximation and expanding :

$$\boxed{\delta W_{\text{STR}} = -\delta \underline{\Phi}^T \underline{\mathbf{K}}_{\text{STR}}^{(N)} \underline{\Phi}}, \quad (35)$$

$$\underline{\mathbf{K}}_{\text{STR}}^{(N)} = \begin{bmatrix} \underline{\mathbf{K}}_{\text{STR},11}^{(N)} & \underline{\mathbf{K}}_{\text{STR},12}^{(N)} \\ \underline{\mathbf{K}}_{\text{STR},21}^{(N)} & \underline{\mathbf{K}}_{\text{STR},22}^{(N)} \end{bmatrix} = \begin{bmatrix} \underline{\mathbf{N}}_w(\eta_B) K_{\text{STR}} \underline{\mathbf{N}}_w^T(\eta_B) & -x_B K_{\text{STR}} \underline{\mathbf{N}}_w(\eta_B) \underline{\mathbf{N}}_\theta^T(\eta_B) \\ -x_B K_{\text{STR}} \underline{\mathbf{N}}_\theta(\eta_B) \underline{\mathbf{N}}_w^T(\eta_B) & x_B^2 K_{\text{STR}} \underline{\mathbf{N}}_\theta(\eta_B) \underline{\mathbf{N}}_\theta^T(\eta_B) \end{bmatrix} \quad (36)$$

Principle of virtual work and convergence criterion

Assembling all contributions, $\underline{\mathbf{K}}_S^{(N)} = \underline{\mathbf{K}}_{\text{WSTIF}}^{(N)} + \underline{\mathbf{K}}_{\text{STR}}^{(N)}$, the aeroelastic eigenproblem reads

$$\left[\underline{\mathbf{K}}_S^{(N)} - q \underline{\mathbf{K}}_A^{(N)} \right] \underline{\Phi} = \underline{\mathbf{0}}. \quad (37)$$

The divergence pressure and convergence indicator are

$$\boxed{q_D^{(N)} = \min \left\{ q > 0 \mid \det(\underline{\mathbf{K}}_S^{(N)} - q \underline{\mathbf{K}}_A^{(N)}) = 0 \right\}}, \quad \varepsilon_N = \left| \frac{q_D^{(N)} - q_D^{(N-1)}}{q_D^{(N-1)}} \right|. \quad (38)$$

Rapid stabilization of $q_D^{(N)}$ with N validates the polynomial Ritz basis.